

S.M. NABIL MUSHFIOUE

PORTFOLIO DOSSIER

VERSION 2026.1 • PRODUCTION READY

github.com/nabilmushfique460

FULL-STACK PYTHON & DJANGO SOFTWARE ENGINEER

Executive Profile: High-impact Software Engineer specializing in scalable full-stack web architectures, robust Python backend engineering, Django ecosystem development, RESTful microservices, and database performance tuning. Proven track record delivering **20+ production-grade software projects** spanning enterprise web platforms, headless automated ETL pipelines, computer vision systems, and native cross-platform desktop administrative suites.

20+	100%	6	0ms
PRODUCTION APPS	TESTED ARCHITECTURE	CORE DOMAIN AREAS	SECURITY COMPLIANCE

Core Technical Stack & Engineering Competencies

Backend & Web Frameworks Django 6.x, Django REST Framework, Flask, Python 3.12+, ASGI/WSGI, Gunicorn, Celery, Whitenoise	Frontend & UI/UX Vanilla JavaScript (ES6+), HTML5, CSS3 / Modern Flex & Grid, Responsive UI, Glassmorphism, Micro-interactions
Databases & Caching PostgreSQL, MySQL, SQLite3, Django ORM, 3NF Normalization, Index Optimization, Redis Caching	AI, Vision & Automation OpenCV, NumPy, Matplotlib, BeautifulSoup4, Selenium, OpenPyXL, ReportLab PDF, NLP APIs
Desktop & Native GUI PyQt6, PySide, Qt Designer, Cross-Platform Desktop Packaging (Windows/macOS/Linux), Local DB Engines	DevOps & Best Practices Docker, Docker Compose, Git / GitHub CI/CD, Linux Server Admin, SMTP Email Routing, PEP 8, Clean Architecture

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Professional Services & Architectural Solutions

High-performance technical services delivered with strict quality benchmarks and domain-driven design.

[01] FULL STACK ARCHITECTURE — Custom Full Stack Web Applications

End-to-end bespoke web platforms architected with Django, Python, and modern responsive frontends. Built with domain-driven design, robust ORM modeling, role-based access control (RBAC), secure authentication flows, and high-performance server-side rendering.

Deliverables: • Modular Django / Flask Core Engine • Dynamic Responsive UI with Vanilla JS • Role-Based Permissions & Auth Systems • Unit & Integration Test Coverage

Technologies: Django • Python • HTML5 / CSS3 • JavaScript • SQLite / PostgreSQL

[02] BACKEND & MICROSERVICES — High Throughput REST APIs & Microservices

Stateless, scalable RESTful API backends and webhook microservices built with Django REST Framework and Flask. Engineered for low-latency JSON delivery, granular token authentication (JWT / Bearer), rate-limiting, and comprehensive Swagger/OpenAPI documentation.

Deliverables: • Stateless REST Microservice Endpoints • JWT & Token-Based Authentication • Automated OpenAPI / Swagger Interactive Docs • Rate Limiting, Throttling & Caching

Technologies: Django REST Framework • Flask • JSON API • Redis Caching • Postman

[03] DATABASE & STORAGE — Relational Database Architecture & Query Tuning

High-integrity relational schema design, 3NF normalization, index optimization, and transaction management. Specialized in tuning slow database queries, eliminating N+1 ORM bottlenecks, managing non-destructive migrations, and ensuring ACID consistency.

Deliverables: • Normalized Relational Schema Architecture • Index Optimization & Query Profiling • ACID Transactions & Row-Level Locking • Non-Destructive Migration Scripts

Technologies: PostgreSQL • MySQL • SQLite • Django ORM • SQL Profiling

[04] DATA EXTRACTION & ETL — Automated Data Pipelines & Web Scraping

Automated data harvesting, cleansing, transformation, and ingestion pipelines. Built with BeautifulSoup, Selenium, and background scheduling to reliably extract unstructured web data, parse dynamic JavaScript pages, validate schemas, and sync cleanly into production databases.

Deliverables: • Headless Multi-Threaded Scraping Engines • Anti-Bot Handling & Proxy Support • Schema Validation & ETL Normalization • Automated Scheduled Execution

Technologies: Python • BeautifulSoup • Selenium • Pandas ETL • Cron Automation

[05] COMPUTER VISION & AI — Computer Vision & Intelligent Image Systems

Real-time video stream processing, motion detection, contour analysis, and automated image transformation pipelines using OpenCV and NumPy. Developing computer vision tools that integrate seamlessly with web services and standalone automated workflows.

Deliverables: • Live Webcam Frame Analysis & Motion Tracking • Contour Detection & Visual Filtering • Automated Batch Image Processing • Matplotlib & OpenCV Stream Integration

Technologies: OpenCV • Python • NumPy • Computer Vision • Matplotlib

[06] DESKTOP APPLICATIONS — Cross-Platform Desktop GUI & Enterprise Tools

Native graphical user interface desktop applications built with PyQt6 / PySide. Tailored for administrative dashboards, offline database tools, school and student management portals, automated PDF invoice generation, and custom internal enterprise software.

Deliverables: • Native Cross-Platform GUI (Win/Mac/Linux) • Embedded SQLite Local Database • Automated PDF & Excel Export Engines • Standalone Executable Packaging

Technologies: PyQt6 • Python • Qt Designer • ReportLab PDF • SQLite

Projects Catalog — Web Applications, APIs & Automation

Comprehensive catalog of implemented, production-tested software systems (Projects 01 to 10 of 20).

#01 Django Task Flow & Management Platform

Django & Python | Web Application

Architected a full-lifecycle Django web application featuring robust task creation, multi-attribute status pipelines, dynamic filtering, deadline notifications, and responsive UI components with persistent SQLite database storage.

Stack: Django • Python • Full-Stack • Task Flow • **Repo:** github.com/nabilmushfique460

#02 Responsive Python Portfolio & Showcase Engine

Full-Stack Python | Web Application

Engineered a high-performance personal portfolio platform utilizing dynamic CSV ingestion, customized Django template inheritance, responsive CSS grid layouts, and active SMTP email communication pipelines.

Stack: Python • Vanilla JS • Responsive UI • **Repo:** github.com/nabilmushfique460

#03 Automated PDF Document & Invoice Generator

PDF Automation | Automation

Built an automated document compilation engine leveraging ReportLab and FPDF to programmatically assemble structured business invoices, analytical reports, and data certificates from dynamic input datasets.

Stack: Python • PDF Generator • Document Engine • **Repo:** github.com/nabilmushfique460

#04 Excel Data Pipeline & Spreadsheet Compiler

Excel & PDF Engine | Automation

Constructed an automated spreadsheet processing script using OpenPyXL to ingest multi-sheet tabular workbooks, calculate aggregate metrics, perform structural data validations, and render PDF summaries.

Stack: OpenPyXL • PDF Toolkit • ETL Process • **Repo:** github.com/nabilmushfique460

#05 Real-Time News Sentiment & NLP Analyzer

Sentiment NLP | AI & Data

Engineered an AI-assisted market intelligence script that queries live global news APIs, processes article text using natural language sentiment classification algorithms, and visualizes market tone distributions.

Stack: NLP Engine • News API • Sentiment AI • **Repo:** github.com/nabilmushfique460

#06 Flask Weather Telemetry & Forecast Microservice

Flask Microservice | REST API

Deployed a lightweight Flask microservice providing structured RESTful endpoints for meteorological forecasting, integrating with remote weather APIs, and formatting serialized JSON response payloads.

Stack: Flask • RESTful API • JSON Feed • **Repo:** github.com/nabilmushfique460

#07 Hotel & Hospitality Reservation Engine

Booking Engine | Web Application

Developed a comprehensive room reservation and hospitality booking interface with date-range room availability verification, customer confirmation emails, and administrative occupancy monitoring.

Stack: Python • Booking Architecture • UX Flow • **Repo:** github.com/nabilmushfique460

#08 Headless Event Scraper & Instant Email Alerter

Web Scraping | Automation

Authored an automated web scraping daemon using BeautifulSoup that monitors target concert and conference websites, detects newly published tour dates, and triggers instant automated SMTP email notifications.

Stack: BeautifulSoup • SMTP Mailer • Event Monitor • **Repo:** github.com/nabilmushfique460

#09 Domain Assistant & Conversational Knowledge Agent

Domain Assistant | AI & Data

Built an interactive question-answering assistant that indexes domain knowledge documents, accepts natural language queries from terminal or web, and provides contextualized synthesized responses.

Stack: Knowledge Base • AI Agent • NLP Queries • **Repo:** github.com/nabilmushfique460

#10 Social Messaging Bot & Conversational Gateway

Conversational AI | [AI & Data](#)

Integrated a webhook-driven conversational bot with the Facebook Graph API to automate customer inquiry handling, recognize intent patterns, and provide automated multi-turn customer support.

Stack: Facebook API • NLP Engine • Chat Gateway • **Repo:** github.com/nabilmushfique460

Projects Catalog — Vision, Analytics, Desktop & E-Commerce

Comprehensive catalog of implemented, production-tested software systems (Projects 11 to 20 of 20).

#11 Computer Vision Security & Motion Detection System

[Computer Vision](#) | [AI & Data](#)

Architected a real-time computer vision security script utilizing OpenCV to analyze live camera frames, calculate background differential contours, detect unauthorized motion, and transmit captured photo frames via automated email.

Stack: OpenCV • Motion Detection • Email Trigger • **Repo:** github.com/nabilmushfique460

#12 System Keystroke Telemetry & Productivity Analytics

[Keystroke Analytics](#) | [Analytics](#)

Built a background operating system telemetry daemon that records typing frequency, performs word count aggregations, and produces analytical productivity metrics without degrading user latency.

Stack: Keyboard Hook • Word Analytics • Stats Daemon • **Repo:** github.com/nabilmushfique460

#13 Network Traffic Telemetry & Real-Time Grapher

[Network Telemetry](#) | [Analytics](#)

Implemented a network telemetry tool monitoring real-time packet ingress/egress, logging historical bandwidth utilization metrics, and rendering interactive web charts using Matplotlib and Bokeh.

Stack: Network Telemetry • Real-Time Graph • Web Plotter • **Repo:** github.com/nabilmushfique460

#14 Distributed Server Health Monitoring Daemon

[Monitoring Daemon](#) | [Analytics](#)

Constructed an automated server health monitor running scheduled background cron jobs to ping remote HTTP endpoints, verify database availability, and dispatch alert messages upon service degradation.

Stack: Remote Server • Cron Monitor • Health Check • **Repo:** github.com/nabilmushfique460

#15 Data Science Analytics Pipeline & Visual Dashboard

[Data Visualization](#) | [Analytics](#)

Built a comprehensive Python data visualization toolkit ingesting multi-variable CSV datasets, conducting statistical aggregation, and rendering publishable scientific scatter plots and bar distributions.

Stack: Matplotlib • Data Pipeline • Visual Analytics • **Repo:** github.com/nabilmushfique460

#16 5-Day Weather Forecast & Historical Analytics Tool

[Weather Dashboard](#) | [Analytics](#)

Authored an analytical desktop dashboard querying OpenWeatherMap REST endpoints, processing multi-day barometric pressure and temperature time-series arrays, and rendering comparative trend lines.

Stack: 5-Day Forecast • Data Charts • API Client • **Repo:** github.com/nabilmushfique460

#17 Enterprise Academic & Student Management Desktop GUI

[PyQt6 & SQL](#) | [Desktop GUI](#)

Designed a native desktop application in PyQt6 providing student registration, enrollment records management, dynamic search filters, relational SQLite persistence, and instant grade transcript generation.

Stack: PyQt6 Desktop • SQLite DB • Admin Dashboard • **Repo:** github.com/nabilmushfique460

#18 Institutional Operations & Course Administrative Suite

[PyQt6 & SQL](#) | [Desktop GUI](#)

Created a standalone administrative desktop suite built with PySide/PyQt6 enabling staff to organize academic departments, manage faculty assignments, and export comprehensive audit records.

Stack: PyQt6 Desktop • Relational SQL • Academic Ops • **Repo:** github.com/nabilmushfique460

#19 Django Dynamic Restaurant Ordering & Menu Engine

[Django Web App](#) | [Web Application](#)

Engineered a responsive Django food ordering platform with real-time category filtering, ingredient customization, dynamic shopping cart session state, and order summary generation.

Stack: Django • Real-Time Menu • Dynamic Filters • **Repo:** github.com/nabilmushfique460

[Django E-Commerce](#) | [Web Application](#)

#20 Django Multi-Vendor E-Commerce Platform

Developed a full-featured e-commerce platform incorporating product catalog management, seller dashboards, customer shopping cart workflows, secure checkout processing, and automated transactional receipts.

Stack: Django E-Commerce • Order Processing • Seller Portal • **Repo:** github.com/nabilmushfique460

Architectural Standards & Engagement Protocol

Enterprise-grade engineering methodologies, deployment pipelines, and collaboration workflows.

1. Clean Code & Modular Architecture Every Django app is strictly partitioned into domain-specific modules with explicit service layers, custom managers, decoupled forms, and encapsulated business logic avoiding fat views or bloated models.	2. Rigorous Security Standards Protection against CSRF, SQL Injection, and XSS. Implemented strict environment variable segregation (.env), secure cookie policies, role-based access control (RBAC), and sanitization filters.
3. Scalable Database Engineering Normalized relational schemas (PostgreSQL / MySQL / SQLite), optimized index planning, transaction atomicity, query count profiling, and elimination of N+1 ORM bottlenecks via select_related / prefetch_related.	4. Automated Testing & Continuous Integration Comprehensive unit and integration test coverage using Django TestCase and PyTest. Verified schema migrations, deterministic data fixtures, and automated GitHub Actions CI/CD workflows.

Client Engagement & Project Delivery Phases

Phase 1: Discovery & Architecture Requirement mapping, schema design, API contract definition, and milestone scoping.	Phase 2: Sprint Development Agile milestone iterations, core backend logic, frontend UI synthesis, and database modeling.	Phase 3: QA, Hardening & Delivery Stress testing, security review, Dockerization, documentation, and production deployment.
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Ready to Build or Scale Your Software Platform?
Available for full-stack web application development, custom API microservices, database performance consulting, and bespoke Python software automation.

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Website: Django Portfolio Live Instance (Local / Production)